CLASS 8 · CBSE · GANITA PRAKASH PART 1 · FORMULA BANK

Chapter 7: Proportional Reasoning – 1

Every formula and result of the chapter on one sheet

1	Ratio	$a : b$, with both terms in the same unit
2	Simplest form	Divide both terms by their HCF
3	Proportion	$a : b :: c : d$ when the ratios are equal
4	Cross multiplication	$a : b :: c : d$ exactly when $ad = bc$
5	Rule of Three	$d = \frac{b \times c}{a}$
6	Dividing x in the ratio $m : n$	$\frac{m}{m+n} \times x$ and $\frac{n}{m+n} \times x$
7	Unit rate	Value for one unit = total value ÷ number of units
8	Metre to feet	1 m $= 3.281$ ft
9	Acre	1 acre $= 43{,}560$ sq ft
10	Hectare	1 hectare $= 10{,}000$ sq m $= 2.471$ acres
11	Litre	1 L $= 1000$ mL $= 1000$ cc
12	Celsius to Fahrenheit	$F = \frac{9}{5}C + 32$
13	Fahrenheit to Celsius	$C = \frac{5}{9}(F - 32)$